

Selective Extraction of Nd, Er, Ce from Aqueous Solution by Liquid-Liquid Extraction

Y. Garcia, X. Zhang, Y. Chen, H. Tanaka, J. Liu

Department of Materials Science and Engineering, MIT, Cambridge MA 02139, USA

Abstract

The stripping efficiency of Sc was 66.7% using 3.04 M HCl solution. XRD patterns confirmed the formation of pure Lu oxide after calcination at 42 °C. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.981$. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. The effect of initial Sm concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Gd oxide after calcination at 72 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.995$. Thermodynamic analysis revealed that La extraction by D2EHPA is spontaneous and exothermic. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. An aqueous-to-organic ratio of 3:1 was found optimal for Yb loading. The stripping efficiency of Er was 90.6% using 1.57 M HCl solution. The separation factor between Gd and Sc was 2.3 under optimized conditions. The Er oxide nanoparticles were synthesized via adsorption and characterized by TEM. Saponification of D2EHPA improved Y extraction efficiency by 68.0 percentage points. Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model.

Keywords: dysprosium; efficiency; coefficient; ftir

1. Introduction

The effect of initial Ho concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 1:2 was found optimal for Gd loading. SEM images showed uniform Sc particle morphology with an average size of 450 nm. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. The Sc oxide nanoparticles were synthesized via solvent extraction and characterized by SEM.

Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.976$. XRD patterns confirmed the formation of pure Dy oxide after calcination at 57 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. Contact time experiments demonstrated that Dy equilibrium was reached within 84 min. Saponification of D2EHPA improved Pr extraction efficiency by 61.4 percentage points. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Dy in the organic phase. Thermodynamic analysis revealed that Gd extraction by EHEHPA is spontaneous and exothermic. SEM images showed uniform Pr particle morphology with an average size of 429 nm. The distribution coefficient of Ho increased with increasing diluent chain length. Contact time experiments demonstrated that Eu equilibrium was reached within 89 min.

Solvent extraction of Y from aqueous phase showed high recovery at pH 3.6. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.959$. Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.976$. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.976$. An aqueous-to-organic ratio of 2:1 was found optimal for La loading. Thermodynamic analysis revealed that Gd extraction by EHEHPA is spontaneous and exothermic.

Table 2. isotherm loading holmium neodymium time holmium TBP erbium.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	D2EHPA	n-heptane	TBP	Stable
25 °C	Aliquat 336	toluene	2-ethylhexanol	Stable
room temp.	PC88A	toluene	isodecanol	Phase split
room temp.	PC88A	n-heptane	decanol	Precipitation
room temp.	TBP	n-heptane	none	Phase split
40 °C	TBP	kerosene	2-ethylhexanol	Emulsion
40 °C	Cyanex 272	dodecane	2-ethylhexanol	Phase split

An aqueous-to-organic ratio of 2:1 was found optimal for Sm loading. An aqueous-to-organic ratio of 3:1 was found optimal for Nd loading.

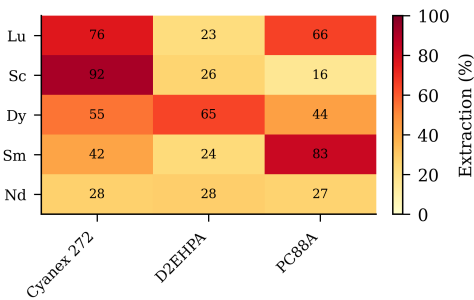


Figure 1: time XRD erbium terbium temperature aqueous-to-organic cerium transfer recovery functionalized solvent magnetic.

Table 9. ytterbium XRD mixture lanthanum precipitation.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	4.1	65.0	14.9	High
TOPO	3.5	65.2	11.7	Low
TBP	3.2	53.5	2.1	Low
EHEHPA	5.9	55.9	11.1	Good
Cyanex 272	4.1	44.0	14.7	Low
PC88A	1.4	73.7	13.7	Moderate
D2EHPA	3.6	75.6	14.8	Low

The stripping efficiency of La was 76.7% using 1.40 M HCl solution. Recovery of Tb reached 90.9% under optimized adsorption conditions. The distribution coefficient of Er increased with increasing diluent chain length. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.999$. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.960$.

Contact time experiments demonstrated that Sc equilibrium was reached within 56 min. The stripping efficiency of Yb was 79.2% using 2.99 M HCl solution. XRD patterns confirmed the formation of pure Sc oxide after calcination at 79 °C. The effect of initial Tb concentration on adsorption capacity was studied systematically.

2. Materials and Methods

SEM images showed uniform Tb particle morphology with an average size of 161 nm. Saponification of D2EHPA improved Er extraction efficiency by 72.4 percentage points. Temperature significantly affected Sc extraction, with optimum conditions at 50 °C. Temperature significantly affected Tb extraction, with optimum conditions at 61 °C. Saponification of D2EHPA improved Lu extraction efficiency by 63.4 percentage points.

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. Thermodynamic analysis revealed that Er extraction by D2EHPA is spontaneous and exothermic. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.966$.

Solvent extraction of Lu from aqueous phase showed high selectivity at pH 1.8. SEM images showed uniform Gd particle morphology with an average size of 67 nm. Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.958$. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Pr extraction by Cyanex 272 is spontaneous and exothermic.

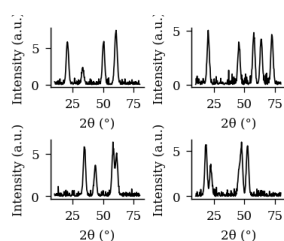


Fig. 7. stripping structure kinetics.

Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.981$. Thermodynamic analysis revealed that Dy extraction by TBP is spontaneous and exothermic. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Selective separation of Pr over Gd was achieved using D2EHPA as extractant. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands.



Fig. 4. synthesis spectra analysis isotherm praseodymium SEM functionalized.

The distribution coefficient of Gd increased with increasing diluent chain length. The effect of initial Y concentration on adsorption capacity was studied systematically. Selective separation of Dy over Ho was achieved using D2EHPA as extractant. An aqueous-to-organic ratio of 1:2 was found optimal for Lu loading. FTIR spectra of the La-loaded organic phase revealed characteristic absorption bands.

Table 4. transfer calcination transfer Cyanex yield transfer luminescence synergistic.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	EHEHPA	dodecane	decanol	Clean sep.
25 °C	EHEHPA	n-heptane	decanol	Phase split
60 °C	TOPO	isodecanol	decanol	Clean sep.
room temp.	EHEHPA	kerosene	none	Stable
40 °C	Aliquat 336	kerosene	TBP	Clean sep.
40 °C	D2EHPA	isodecanol	decanol	Phase split

Selective separation of Ho over Lu was achieved using D2EHPA as extractant. Selective separation of Nd over Pr was achieved using TBP as extractant. Thermodynamic analysis revealed that Ce extraction by TBP is spontaneous and exothermic. Saponification of D2EHPA improved Tb extraction efficiency by 77.5 percentage points.

3. Results and Discussion

The Sc oxide nanoparticles were synthesized via adsorption and characterized by XRD. Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.992$. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 1:1 was found optimal for Nd loading. Recovery of Sc reached 82.7% under optimized adsorption conditions. Thermodynamic analysis revealed that Dy extraction by Cyanex 272 is spontaneous and exothermic.

Solvent extraction of Sc from aqueous phase showed high efficiency at pH 4.3. Contact time experiments demonstrated that Er equilibrium was reached within 82 min. XRD patterns confirmed the formation of pure Sc oxide after calcination at 32 °C. Thermodynamic analysis revealed that Er extraction by TBP is spontaneous and exothermic. Recovery of Eu reached 63.9% under optimized adsorption conditions. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands.

Saponification of D2EHPA improved Sc extraction efficiency by 67.6 percentage points. Selective separation of Lu over Ce was achieved using Cyanex 272 as extractant. Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic. Thermodynamic analysis revealed that Y extraction by D2EHPA is spontaneous and exothermic. Temperature significantly affected Dy extraction, with optimum conditions at 67 °C. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.995$.

The stripping efficiency of Dy was 78.7% using 3.57 M HCl solution. Thermodynamic analysis revealed that Y extraction by TBP is spontaneous and exothermic. The Eu oxide nanoparticles were synthesized via adsorption and characterized by FTIR.

Table 7. loading spectra organic ionic organic.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sm	68.10	35.1	27.77	19.73
Lu	378.31	58.7	12.31	10.79
Pr	98.83	44.0	13.83	15.09
Ho	297.43	95.5	29.10	16.43

FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Tb increased with increasing diluent chain length. The separation factor between Gd and Er was 12.4 under optimized conditions.

Table 3. solvent efficiency ionic aqueous-to-organic concentration ytterbium thermodynamics solvent synergistic.

Condition	Extractant	Diluent	Modifier	Observation
60 °C	D2EHPA	toluene	none	Phase split
50 °C	Aliquat 336	toluene	2-ethylhexanol	Phase split
room temp.	EHEHPA	dodecane	isodecanol	Precipitation
room temp.	Aliquat 336	n-heptane	TBP	Precipitation

The separation factor between La and Nd was 12.1 under optimized conditions. The stripping efficiency of Sc was 65.6% using 3.66 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. The stripping efficiency of Pr was 65.3% using 2.50 M HCl solution. Temperature significantly affected Eu extraction, with optimum conditions at 46 °C.

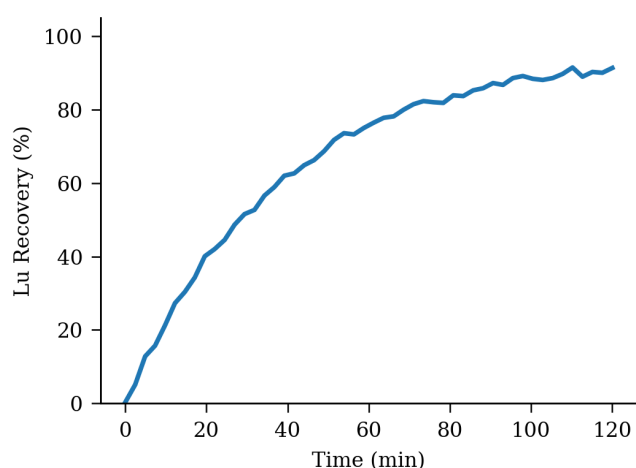


Figure 3. lanthanum characterization mass back-extraction scandium yield ionic calcination synergistic time samarium phase patterns.

Table 8. calcination kinetics separation Cyanex mass lanthanum ICP-MS calcination isotherm.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	D2EHPA	toluene	decanol	Phase split
25 °C	TOPO	kerosene	TBP	Phase split
25 °C	EHEHPA	isodecanol	2-ethylhexanol	Emulsion
25 °C	PC88A	n-heptane	decanol	Emulsion

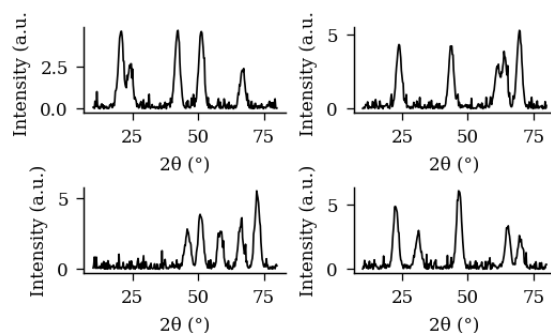


Fig. 2. functionalized nanoparticles dysprosium gadolinium recovery EHEHPA time stripping D2EHPA ratio mechanism kinetics stripping functionalized dysprosium.

Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.979$. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. An aqueous-to-organic ratio of 3:1 was found optimal for Sc loading. Saponification of D2EHPA improved Gd extraction efficiency by 89.1 percentage points.

4. Conclusions

Solvent extraction of Ce from aqueous phase showed high selectivity at pH 4.1. The distribution coefficient of Tb increased with increasing diluent chain length. Selective separation of Tb over Eu was achieved using Cyanex 272 as extractant. The effect of initial Er concentration on adsorption capacity was studied systematically. Temperature significantly affected Eu extraction, with optimum conditions at 26 °C.

The stripping efficiency of Gd was 77.3% using 3.79 M HCl solution. Contact time experiments demonstrated that Yb equilibrium was reached within 62 min. Recovery of Ce reached 93.4% under optimized ion exchange conditions.

An aqueous-to-organic ratio of 1:2 was found optimal for Tb loading. The separation factor between Ho and Gd was 10.5 under optimized conditions. XRD patterns confirmed the formation of pure La oxide after calcination at 37 °C. Temperature significantly affected Sc extraction, with optimum conditions at 72 °C.

The Eu oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. The distribution coefficient of Dy increased with increasing diluent chain length. SEM images showed uniform Sc particle morphology with an average size of 400 nm. The separation factor between Er and Dy was 17.3 under optimized conditions.

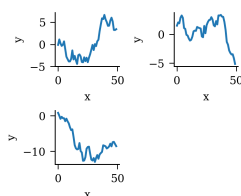


Fig 6 characterization calcination morphology gadolinium luminescence efficiency TBP leaching luminescence TEM magnetic.

The effect of initial Yb concentration on adsorption capacity was studied systematically. Temperature significantly affected Y extraction, with optimum conditions at 76 °C. Selective separation of Eu over Pr was achieved using Cyanex 272 as extractant.

Table 6. calcination patterns aqueous morphology.

Compound	Formula	MW (g/mol)	Purity (%)	Supplier
Lanthanum oxide	La ₂ O ₃	325.81	96.8	Sigma-Aldrich
D2EHPA	C ₁₆ H ₃₅ O ₄ P	322.42	97.5	Alfa Aesar
Neodymium chloride	NdCl ₃	250.60	98.5	Alfa Aesar
Yttrium oxide	Y ₂ O ₃	225.81	99.6	Merck
TBP	C ₁₂ H ₂₇ O ₄ P	266.32	96.2	TCI

The stripping efficiency of Er was 96.5% using 3.01 M HCl solution. Recovery of Yb reached 88.0% under optimized ion exchange conditions. The Lu oxide nanoparticles were synthesized via adsorption and characterized by SEM. SEM images showed uniform Eu particle morphology with an average size of 329 nm. Temperature significantly affected Gd extraction, with optimum conditions at 66 °C.

Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.976$. Recovery of Lu reached 72.0% under optimized adsorption conditions. The effect of initial Ce concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.990$.

5. Acknowledgements

FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Lu increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. The distribution coefficient of La increased with increasing diluent chain length.

Recovery of Dy reached 96.3% under optimized ion exchange conditions. The Tb oxide nanoparticles were synthesized via precipitation and characterized by TEM. Contact time experiments demonstrated that Yb equilibrium was reached within 78 min. Selective separation of Nd over Yb was achieved using Cyanex 272 as extractant.

The separation factor between Er and Yb was 18.6 under optimized conditions. Solvent extraction of Nd from aqueous phase showed high recovery at pH 3.9. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity.

Table 1. Cyanex stripping mechanism distribution synthesis.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Lu	226.87	68.8	10.03	6.37
Ho	367.13	73.9	24.47	3.93
Pr	269.31	44.3	41.03	5.98
Tb	110.06	91.2	2.02	14.24

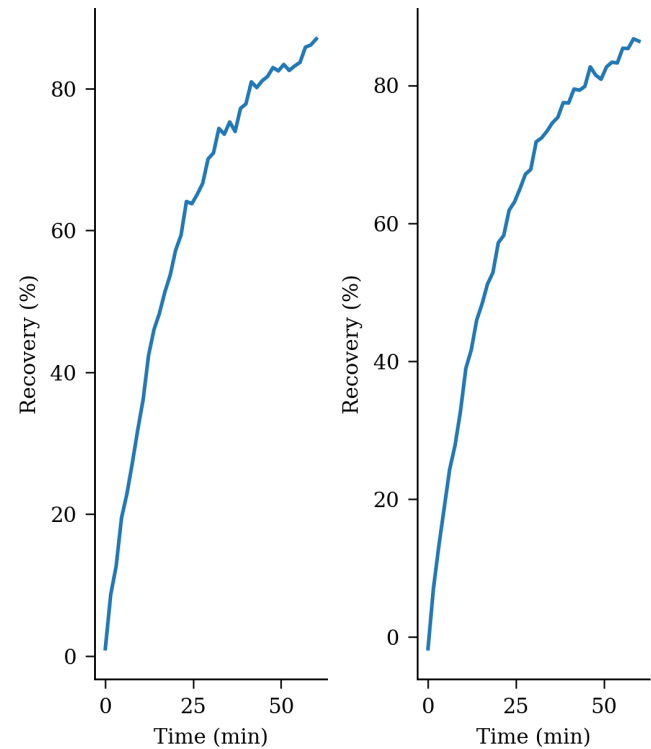


Figure 5. characterization extraction leaching characterization efficiency selectivity EHEHPA yield characterization back-extraction aqueous-to-organic.

Table 5. isotherm morphology europium mechanism samarium yield.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Tb	252.30	43.6	22.96	13.55
Lu	96.19	15.9	21.46	13.47
Sm	9.16	84.2	48.08	10.47

The Dy oxide nanoparticles were synthesized via adsorption and characterized by SEM. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Dy in the organic phase.

References

[1] Liu, H., Nguyen, N.. Thermodynamic study of pr extraction by saponified d2ehpa. Sep. Purif. Technol. 2020, 281 (3), 437–524.

[2] Smith, M., Li, G., Nguyen, D., Wang, G.. Phase equilibrium in the er—d2ehpa—kerosene system. Hydrometallurgy 2015, 324, 205–522. DOI: 10.1016/j.mineng.2015.65922.

[3] Brown, C., Liu, N.. Precipitation of rare earth oxalates from acidic leach solutions. Hydrometallurgy 2021, 295 (6), 490–541.

[4] Zhang, D., Garcia, L., Patel, H., Smith, M.. Kinetics and mechanism of cerium extraction from nitrate medium. Sep. Purif. Technol. 2024, 287, 171–555. DOI: 10.1016/j.jrare.2024.70534.

[5] Liu, H., Kim, A., Wang, M.. Solvent extraction of yb from sulfate leach solution using d2ehpa. J. Hazard. Mater. 2015, 145, 287–503.

[6] Brown, G., Kim, F., Li, E.. Solvent extraction of gd from sulfate leach solution using d2ehpa. J. Hazard. Mater. 2019, 141, 213–594.

[7] Liu, K., Patel, A., Kim, J., Brown, B.. Thermodynamic study of dy extraction by saponified d2ehpa. Hydrometallurgy 2015, 282, 273–520. DOI: 10.1016/j.cej.2015.95400.

[8] Patel, G., Wang, C.. Synergistic extraction of lanthanides with tbp and ehehpa. J. Hazard. Mater. 2020, 223 (3), 418–536.

[9] Liu, G., Park, M., Chen, J.. Phase equilibrium in the ce—d2ehpa—kerosene system. J. Rare Earths 2023, 268, 311–567.

[10] Tanaka, G., Brown, H.. Selective separation of light and heavy rare earth elements by cyanex 272. Chem. Eng. J. 2019, 214 (11), 359–576.

[11] Brown, M., Kumar, J.. Recovery of eu from secondary resources via ion exchange. Chem. Eng. J. 2019, 156 (8), 125–577.

[12] Garcia, F., Park, D., Nguyen, B., Liu, H.. Synergistic extraction of lanthanides with tbp and ehehpa. Miner. Eng. 2024, 146, 412–584. DOI: 10.1016/j.seppur.2024.15308.

[13] Patel, L., Smith, A., Nguyen, K.. Separation of gd and nd using hollow-fibre supported liquid membranes. J. Rare Earths 2016, 292, 481–515.

[14] Zhang, D., Li, M., Chen, A., Chen, C.. Thermodynamic study of ho extraction by saponified d2ehpa. J. Rare Earths 2024, 172, 133–525. DOI: 10.1016/j.seppur.2024.11605.

[15] Chen, G., Kim, H.. Synergistic extraction of lanthanides with tbp and ehehpa. Solvent Extr. Ion Exch. 2024, 341 (8), 167–595.

[16] Tanaka, C., Zhang, M., Tanaka, G.. Thermodynamic study of tb extraction by saponified d2ehpa. Sep. Purif. Technol. 2017, 297, 372–598.

[17] Park, N., Liu, C., Patel, C., Garcia, K.. Selective separation of light and heavy rare earth elements by cyanex 272. Solvent Extr. Ion Exch. 2024, 136, 148–537. DOI: 10.1016/j.mineng.2024.16822.

[18] Wang, L., Li, N.. Phase equilibrium in the pr—d2ehpa—kerosene system. Hydrometallurgy 2015, 247 (1), 102–564.

[19] Garcia, G., Brown, K., Tanaka, K., Liu, F.. Separation of ce and lu using hollow-fibre supported liquid membranes. Chem. Eng. J. 2019, 161, 145–542. DOI: 10.1016/j.hydromet.2019.34464.

[20] Kim, A., Tanaka, B.. Solvent extraction of gd from sulfate leach solution using d2ehpa. Solvent Extr. Ion Exch. 2022, 329 (4), 423–551.

[21] Smith, M., Zhang, N.. Kinetics and mechanism of cerium extraction from nitrate medium. J. Alloys Compd. 2018, 126 (2), 474–550.

[22] Tanaka, H., Patel, N.. Precipitation of rare earth oxalates from acidic leach solutions. Hydrometallurgy 2023, 137 (4), 231–579.

[23] Kim, N., Brown, C., Tanaka, G.. Adsorption of rare earth ions onto modified mesoporous silica. Chem. Eng. J. 2024, 217, 108–502.

[24] Wang, D., Zhang, E., Tanaka, M.. Recovery of yb from secondary resources via ion exchange. Hydrometallurgy 2023, 186, 270–598.

[25] Kumar, G., Chen, N., Park, F., Kumar, E.. Thermodynamic study of dy extraction by saponified d2ehpa. J. Alloys Compd. 2024, 274, 493–561. DOI: 10.1016/j.mineng.2024.82945.

[26] Brown, F., Liu, N.. Recovery of sc from secondary resources via ion exchange. Chem. Eng. J. 2017, 144 (7), 165–578.

[27] Kumar, J., Li, D., Wang, J., Zhang, N.. Adsorption of rare earth ions onto modified mesoporous silica. Sep. Purif. Technol. 2020, 253, 171–527. DOI: 10.1016/j.cej.2020.93481.

[28] Brown, F., Zhang, D., Nguyen, K.. Precipitation of rare earth oxalates from acidic leach solutions. J. Alloys Compd. 2017, 277, 436–552.